

University of Pisa – Data Science and Business Informatics

Cost Foresight: Forecasting Production Costs in Steel

Exploratory Analysis, Feature Engineering and Forecasting Models

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Course: Fundamentals of Business Management

Academic Year: 2025–2026



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Contents

1	Introduction	3
1.1	Context	3
1.2	Report Structure	3
2	Business Case	4
2.1	The Nucor Operating Model and Market Position	4
2.2	The Problem: A Structural Break in Volatility	4
2.3	The Strategic Value of “Cost Foresight”	5
2.4	Project Objectives	5
3	Strategic Data Methodology	5
3.1	Feature Selection via Risk Factor Analysis	5
3.2	Data Engineering and Logic	6
4	Data Description	6
4.1	Data Sources	6
5	Operational and Market Trends Analysis	7
5.1	Descriptive Statistics	7
5.2	Trend Dynamics and Structural Breaks	8
5.3	Segment Analysis: Sales Composition	9
5.4	Financial Performance: Gross Margin	10
5.5	Standardized Deviations and Outlier Detection	11
5.6	Cost–Volume Relationship Analysis	12
5.7	Rolling Correlation Dynamics	13
6	Feature Engineering	14
6.1	Base Feature Transformation	14
6.2	Temporal Feature Construction	14
6.3	Feature Selection Strategy	14
6.3.1	Expanded Feature: Correlation, Lagged Effects, and Rolling Dynamics	14
6.3.2	Statistical Feature Selection for the Non-Lagged Feature Set	16
6.4	Synthetic Data Augmentation	16
7	Modeling Approach	17
7.1	Feature Sets	17
7.2	Cross-Validation Strategy	17
7.3	Baseline Models	17
7.4	Regularized Linear Models	17
7.4.1	Ridge Regression (L_2 Regularization)	18
7.4.2	Lasso Regression (L_1 Regularization)	18
7.5	Random Forest Regression	18
7.6	Experimental: Deep Learning Feasibility	18
8	Results	18
8.1	Performance on the Original Dataset	19
8.2	Effect of Lagged Features	19
8.3	Stress Test: Ukraine-War Volatility Window	19
8.4	Synthetic Data Experiment (Gaussian Copula)	20
8.5	Models Benchmarking with Synthetic Augmentation	20
8.5.1	Naive Benchmark	21
8.5.2	Model Performance	21
8.5.3	Interpretation	21

8.6	Results of Neural Network Models	22
9	Conclusions	23
9.1	Key Findings	23
9.2	Strategic Summary	23
9.3	Benefits Analysis	24
9.4	Future Research Directions	24

Abstract

The steel manufacturing industry faces extreme cost volatility driven by fluctuating raw materials, energy prices, and geopolitical shocks. This report develops a forecasting framework for production costs at Nucor Corporation using quarterly data that combines internal financial disclosures with macroeconomic indicators. Working within a data-scarce environment, we employ synthetic data augmentation through Gaussian Copula modeling and systematically compare baseline heuristics, regularized linear models, and deep learning architectures. The results reveal that parsimonious models substantially outperform complex neural networks in small-sample regimes. Regularized linear approaches offer marginal gains in prediction quality, while deep learning exhibits pronounced overfitting. Feature analysis shows that both macroeconomic variables (such as scrap prices, logistics costs, and broad commodity price indices) and lagged cost dynamics contribute substantially to model performance. While their relative importance varies across models and market conditions, both types of features play a central role in explaining production cost fluctuations. For capital-intensive industries with sparse data availability, the findings indicate that regularized linear estimators offer more stable performance than complex algorithmic alternatives. Synthetic data can support exploratory analysis, but do not replace real temporal information, making interpretability central to effective decision support.

1 Introduction

Forecasting production costs in the steel manufacturing sector is a complex and strategically critical task. Cost behavior is highly sensitive to fluctuations in raw material prices, energy markets, transportation, and broader macroeconomic forces. Reliable predictions support managerial decisions in pricing, budgeting, procurement, and operational planning.

This report combines internal financial disclosures with external market indicators to evaluate the feasibility of predictive modeling for steel production costs and to identify the primary drivers of cost variability. In addition to forecasting, the analysis examines how internal operational metrics interact with macroeconomic conditions, highlights major deviations associated with market disruptions such as COVID-19 and the war in Ukraine, and compares the stability of different modeling approaches under alternative feature specifications.

1.1 Context

The global steel industry is navigating unprecedented volatility. In 2024, global crude steel production reached approximately **1.88 billion tonnes**, yet the market faces a looming overcapacity crisis. The OECD projects [5] that an additional **165 million tonnes** of capacity will come online between 2025 and 2027. This combination of supply saturation and geopolitical instability—intensified by the war in Ukraine and post-COVID supply chain shocks—has made accurate cost prediction a critical competitive differentiator.

For **Nucor Corporation**, which shipped approximately **24.8 million tons** of steel to outside customers in 2024, navigating this environment is essential. Unlike competitors reliant on blast furnaces, Nucor operates using Electric Arc Furnaces (EAF), a more flexible but highly market-sensitive technology that exposes production costs to immediate movements in scrap and electricity prices. In this context, the *“Cost Foresight”* project leverages machine learning to forecast Nucor’s **Cost of Goods Sold (COGS) per Ton** one quarter ahead, replacing static historical averaging with a dynamic, data-driven approach.

1.2 Report Structure

The report is organized as follows:

- **Section 2:** Business case and motivation
- **Section 3:** Strategic Data Methodology
- **Section 4:** Dataset Description
- **Section 5:** Operational and Market Trends Analysis

- **Section 6:** Feature Engineering
- **Section 7:** Modeling Approaches
- **Section 8:** Results and Interpretation
- **Section 9:** Conclusions and Future Work

2 Business Case

The steel industry faces one of the most volatile cost environments in manufacturing. Production costs fluctuate rapidly in response to scrap metal prices, energy costs, and transportation dynamics, while global macroeconomic shocks can trigger abrupt operational challenges.

Managing these costs is essential for maintaining profitability and ensuring operational efficiency. Even short-term volatility can significantly affect quarterly margins and planning decisions. This context motivates the present study, which aims to quantify cost drivers and assess forecasting approaches.

2.1 The Nucor Operating Model and Market Position

Nucor, the largest steel producer in North America, recycles approximately 18 million gross tonnes of scrap steel annually. Its business model relies heavily on the spread between finished steel prices and raw material costs. This spread tightened substantially in 2024: while Nucor’s average sales price per ton decreased by 10%, the prices of scrap and scrap substitutes declined by only 6% (to roughly \$394 per gross ton). This margin compression underscores the need for accurate and responsive cost forecasting.

2.2 The Problem: A Structural Break in Volatility

Historical budgeting frameworks typically assume smooth, linear cost evolution. However, industry data indicate a clear *structural break* beginning in 2021 (see Figure 1):

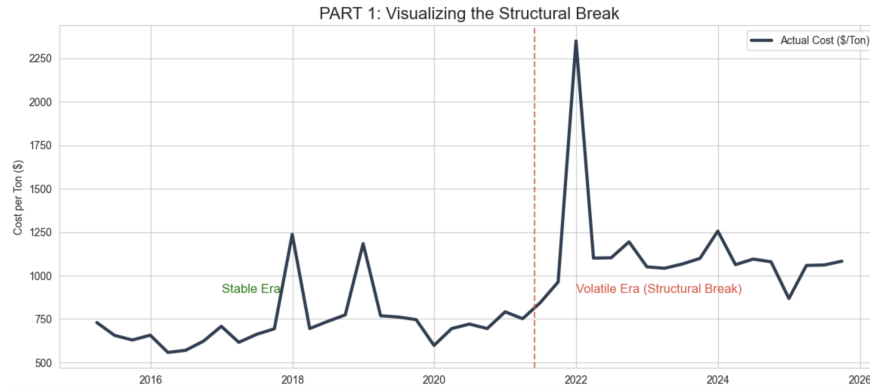


Figure 1: Structural break in 2021

- **Price Volatility:** Global steel prices surged to historic highs (above \$1,100/tonne) in 2021–2022 before correcting, establishing a new regime of amplified price swings.
- **Scrap Sensitivity:** Recent monthly declines in scrap prices exceeding 5–10% make static quarterly budgets unreliable within weeks.
- **Operational Scale:** For an annual production level of approximately 25 million tons, an error of just \$10 per ton corresponds to \$250 million variance in projected Cost of Goods Sold (COGS).

2.3 The Strategic Value of “Cost Foresight”

The proposed model directly addresses this \$250 million exposure. By forecasting production costs one quarter ahead, Nucor can:

1. **Hedge Aggressively:** Secure bulk scrap or electricity contracts during expected price dips, potentially saving millions given an 18-million-ton scrap requirement.
2. **Protect Margins:** Anticipate periods in which costs decrease more slowly than sales prices (such as the 4% gap observed in 2024) and adjust pricing strategies accordingly.
3. **Optimize Inventory:** Adapt throughput across roughly 9.5 million tons of mill capacity to avoid holding high-cost inventory during deflationary cycles.

2.4 Project Objectives

This project addresses two primary objectives:

1. **Understand cost dynamics:** Identify how market indicators interact with internal production and financial metrics, detecting patterns, anomalies, and periods of instability. This understanding is essential for interpreting cost behavior and assessing the company’s economic resilience.
2. **Develop predictive models:** Evaluate whether machine learning models can capture these relationships sufficiently to support forward-looking decision-making, particularly in uncertain environments characterized by supply chain disruptions and geopolitical events.

3 Strategic Data Methodology

To translate Nucor’s business environment into a predictive framework, a data strategy was developed based on financial risk analysis rather than arbitrary feature selection. This approach ensures alignment between the model inputs and the economic exposures explicitly recognized by the company.

3.1 Feature Selection via Risk Factor Analysis

Feature selection was guided by Nucor’s own disclosures in the SEC 10-K, *Item 1A: Risk Factors*. This guarantees that the model incorporates the external macroeconomic conditions that directly affect Nucor’s operating margins, competitive position, and demand outlook.

- **Risk 1: Raw Material Volatility**

Nucor Disclosure: Management identifies “fluctuations in the cost of raw materials and energy” as a primary threat to profitability.

Model Input: Quarterly indices for Scrap Steel (X1), Industrial Electricity (X2), and Natural Gas (X3) were integrated to capture volatility in input costs.

- **Risk 2: Global Trade & Competition**

Nucor Disclosure: The company cites “global trade regulations and competition” as major risks, particularly due to foreign steel dumping and currency effects.

Model Input: The US Dollar Index (X9) and US Scrap Exports (X8) were included to represent currency volatility and the outflow of domestic scrap to international markets.

- **Risk 3: Macroeconomic Uncertainty**

Nucor Disclosure: Demand is characterized as highly cyclical and sensitive to “general economic uncertainty.”

Model Input: The Economic Policy Uncertainty Index (X11) was incorporated to quantify the macro-political climate influencing steel demand and market stability.

3.2 Data Engineering and Logic

Transforming raw internal and external indicators into a cohesive dataset required addressing several structural reporting challenges.

- **Recovering the “Hidden Q4”**

A key obstacle in quarterly modelling is that annual 10-K filings frequently aggregate fourth-quarter results into the full-year total, obscuring Q4-specific outcomes. To resolve this, a reconstruction layer was implemented:

$$Q4 = \text{Annual Total} - (Q1 + Q2 + Q3).$$

This logic produced a continuous, uninterrupted quarterly financial series from 2015 to 2025, restoring missing Q4 statements with validated mathematical consistency.

- **Temporal Alignment and Lag Features**

Commodity price movements (e.g., scrap steel) do not affect Cost of Goods Sold (COGS) immediately; they propagate through inventory cycles with a delay. To capture this accounting mechanism, lagged features were generated (e.g. Scrap Price_{t-1}), along with rolling-window transformations such as moving averages and rolling standard deviations. These features reflect the true temporal structure of cost recognition.

4 Data Description

The analysis leverages Nucor Corporation’s financial and operational data, the largest steel producer in the United States and a pioneer in Electric Arc Furnace (EAF) steelmaking. Nucor reports its financial and operational metrics on a **quarterly basis**, which defines the temporal granularity of our analysis. The final dataset contains **43 reporting periods** spanning **2015–2025** and includes **31 engineered features** derived from both internal SEC filings and external macroeconomic sources. All missing values were resolved through a structured data processing pipeline, resulting in a complete and analysis-ready dataset.

The dataset is designed to model the target variable **Cost per Ton**, representing the precise production cost per ton of steel sold. This metric was either directly extracted from Nucor’s periodic filings or mathematically reconstructed in cases where only annual totals were available.

The data processing workflow resulted in two distinct final datasets, ensuring both granular financial visibility and modeling efficiency:

- **Master Dataset (General):** A comprehensive repository containing 31 variables. It includes all raw financial metrics, SEC filing metadata (e.g., accession numbers, report types), and intermediate calculation fields. This dataset serves as the primary reference point for exploratory analysis and auditability.
- **Master Dataset (Modelling):** A streamlined subset optimized for regression analysis. It retains only the target variable (Y) and the 13 engineered feature vectors ($X_1 - X_{13}$).

4.1 Data Sources

The dataset integrates two primary categories of information, combining internal company disclosures with external macroeconomic indicators.

1. Internal Data (SEC 10-K / 10-Q Filings)

Financial and operational variables were extracted directly from Nucor’s official filings. These include:

- **Cost of Products Sold**
- **Steel Shipments** (outside customers)
- **Inventory Levels**

- **Total Scrap Used**
- **Energy Consumption Metrics**

These variables form the core of the company-specific component of the dataset, capturing Nucor’s operational structure and production cost dynamics.

2. External Macroeconomic and Industry Data

To incorporate broader economic conditions and upstream/downstream factors, we integrate publicly available datasets from authoritative U.S. institutions. Table 1 provides an overview of each feature, its source, and the rationale for inclusion.

Table 1: External Features, Data Sources, and Motivation

Feature	Source	Rationale
Scrap Price	BLS (PPI)	Primary raw material cost for Electric Arc Furnace (EAF) production.
Electricity Price	EIA	Major energy input required for EAF steelmaking.
Natural Gas	EIA	Key feedstock for DRI-based upstream processes.
Diesel Price	EIA	Proxy for trucking and logistics costs.
Rail Freight Price	BLS	Indicator of heavy transport and freight costs.
Disaster Count	NOAA	Captures disruptive “shock” events affecting supply chains.
Labor Wage	BLS	Direct labor cost component.
Scrap Exports	U.S. Census	Reflects global demand pressure on domestic scrap availability.
USD Index	FRED	Measures currency volatility and macroeconomic risk.
Graphite Electrode Price	BLS	Cost of critical EAF consumables (electrodes).
Policy Uncertainty Index	Economic Policy Uncertainty Index	Proxy for tariff, regulatory, and policy instability.
Inventory Turnover	Internal / Engineered	Indicator of operational efficiency and inventory management.

5 Operational and Market Trends Analysis

This section summarizes the key empirical patterns emerging from the dataset, focusing on the joint evolution of production volumes, unit costs, and financial performance metrics over the period 2015-2025. The analysis combines descriptive statistics, trend inspection, and structural indicators designed to capture macroeconomic shocks and internal operational dynamics. Particular attention is devoted to identifying regime changes associated with the COVID-19 pandemic and the Ukraine conflict.

5.1 Descriptive Statistics

Baseline statistics highlight substantial variability in both *Cost per Ton* and *Tons Shipped*. Table 2 presents summary statistics for the key operational metrics.

Table 2: Descriptive Statistics for Key Variables (2015–2025)

Variable	Mean	Std Dev	Min	Max	Range
Cost per Ton (\$/ton)	865.4	309.1	557.7	2349.1	1791.4
Tons Shipped (millions)	6.52	1.24	3.15	8.75	5.60
COGS (\$ billions)	5.64	2.18	3.42	8.71	5.29

Over the sample window, the production cost increases by approximately **48%** (from \$729/ton in 2015 Q1 to \$1,083/ton in 2025 Q4), while shipments grow by about **20%** (from 5.85M to 7.02M tons). The simple correlation between the two variables is moderately negative ($\rho = -0.434$), indicating that higher output volumes tend to coincide with lower unit production costs—a pattern consistent with economies of scale in electric arc furnace (EAF) steel production.

This inverse relationship is visually apparent in Figure 2, which illustrates the contrasting dynamics of the two series over time using min-max normalization to enable direct comparison.

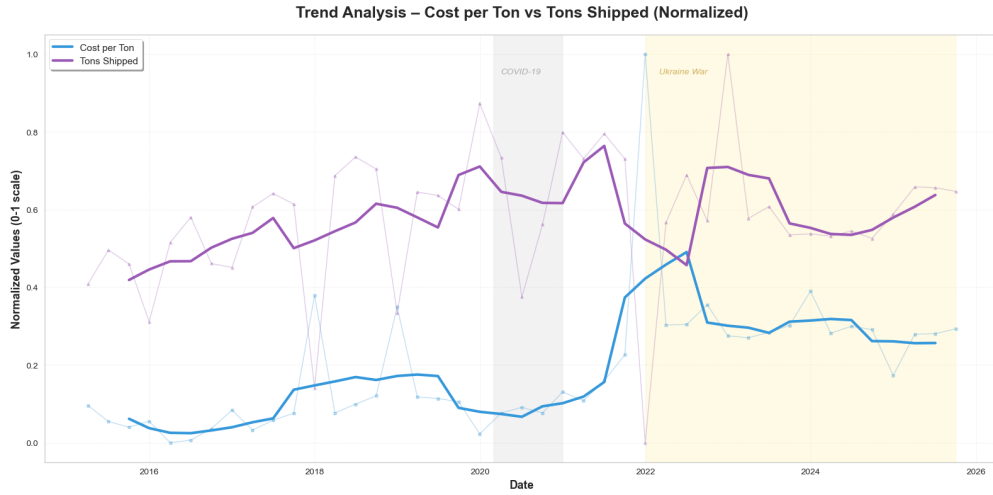


Figure 2: Normalized Trend Analysis — Cost per Ton vs. Tons Shipped (2015–2025).

5.2 Trend Dynamics and Structural Breaks

Normalized trend plots reveal three salient patterns:

1. **Cost volatility dominates:** *Cost per Ton* exhibits pronounced spikes, most notably in 2022 Q2 when it reaches \$2,349/ton (+171% above historical mean), reflecting energy price shocks and supply-chain disruptions following Russia’s invasion of Ukraine.
2. **Volume stability with periodic contractions:** *Tons Shipped* remains relatively stable over time (coefficient of variation = 19%), with temporary contractions around major external events—particularly the 2022 Q1 trough (3.15M tons, -52% below mean), suggesting demand destruction during peak cost periods.
3. **Divergence during crises:** Joint visualization highlights the decoupling of costs and volumes during the Ukraine conflict, where costs rise sharply while volumes temporarily weaken, violating the typical inverse relationship and indicating external shocks overwhelming internal operational dynamics.

Figure 3 provides disaggregated views with dual y-axes, enabling direct interpretation of absolute values alongside normalized trends.

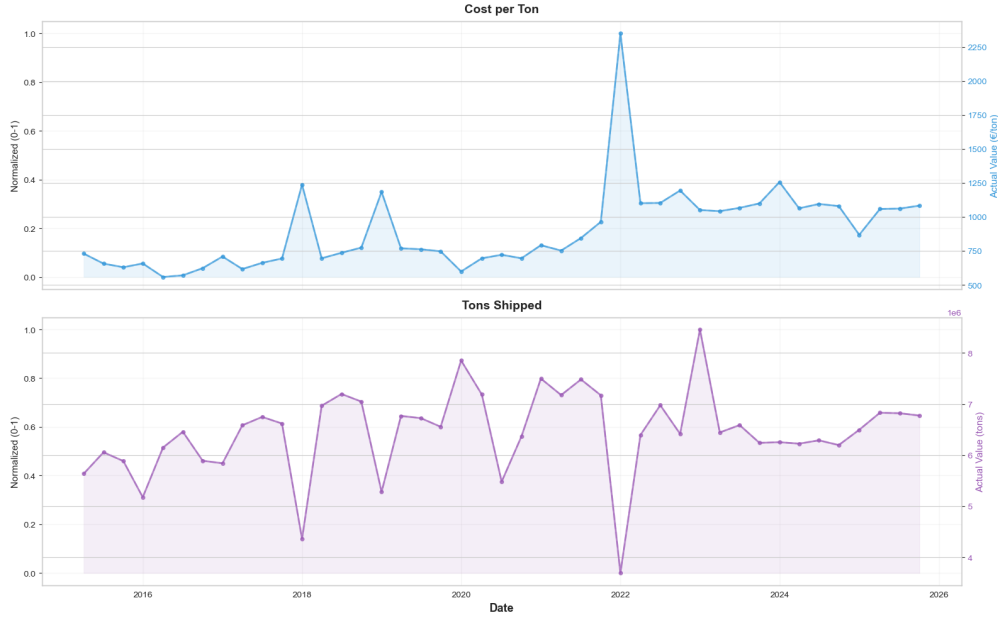


Figure 3: Separated Trend Analysis with Dual Axes. Top panel: Cost per Ton (blue, left axis) with normalized trend (dashed, right axis). Bottom panel: Tons Shipped (purple, left axis) with normalized trend (dashed, right axis).

5.3 Segment Analysis: Sales Composition

To understand the specific drivers of Nucor’s revenue scale, we decomposed Net Sales by product type. Figure 4 illustrates the quarterly performance of Nucor’s three primary reporting segments.

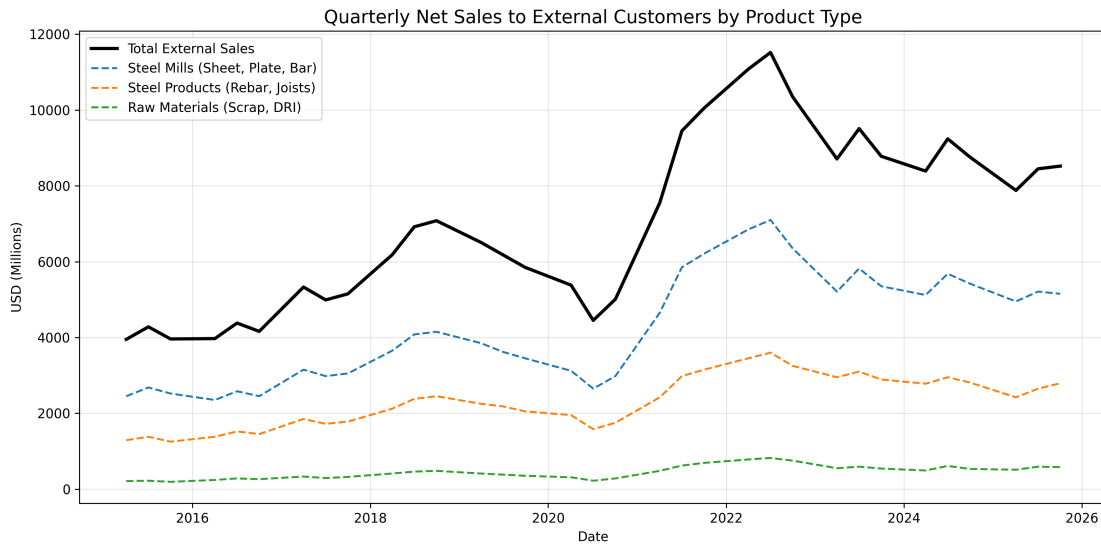


Figure 4: Quarterly Net Sales to External Customers by Product Type (2015–2025). Steel Mills (blue dashed) represent the largest revenue stream, followed by Steel Products (orange dashed) and Raw Materials (green dashed).

The breakdown confirms that Steel Mills (producing sheet, plate, and bar steel) are the primary driver of external revenue, consistently outperforming the downstream Steel Products segment. The Raw Materials

segment contributes a smaller, relatively stable portion of sales. This dominance of the Steel Mills segment implies that Nucor’s overall financial performance is highly sensitive to the cost dynamics of raw steel production, specifically scrap and energy inputs.

5.4 Financial Performance: Gross Margin

While sales volume provides a measure of scale, profitability is determined by the spread between sales price and production costs. Figure 5 tracks Nucor’s consolidated Gross Margin percentage over the 2015-2025 period.

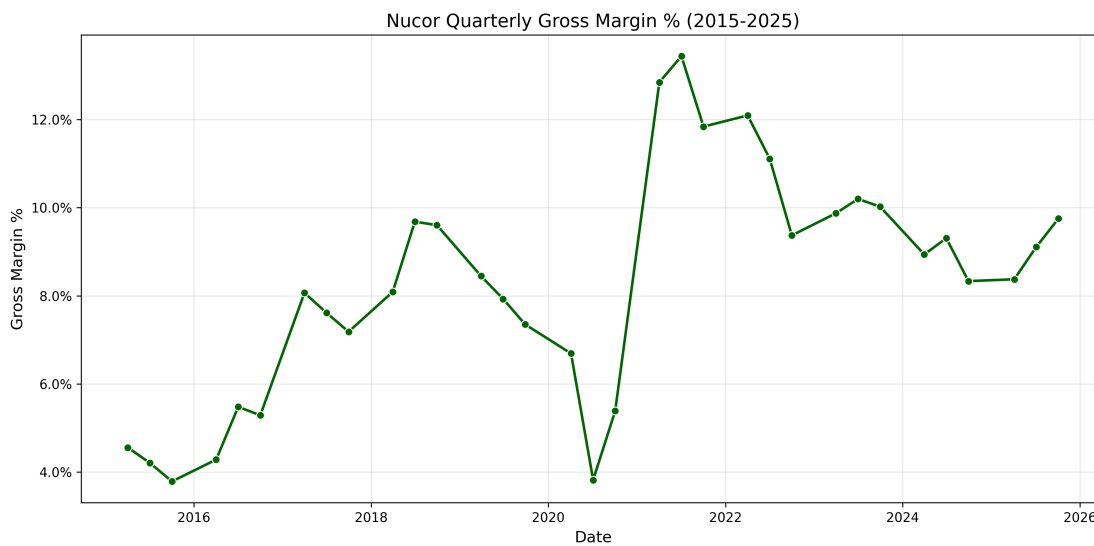


Figure 5: Nucor Quarterly Consolidated Gross Margin % (2015–2025). The trend reveals significant volatility, particularly during the post-2021 inflationary period.

The margin trend highlights the “structural break” in volatility discussed in the Business Case. Pre-2021 margins were relatively stable, whereas the post-2021 era is characterized by sharp expansions (during the post-COVID demand surge) followed by rapid compressions as input costs rose. This volatility reinforces the need for predictive models that can anticipate cost shifts to protect these fluctuating margins.

Table 3 provides a comprehensive annual summary connecting these financial outcomes with operational efficiency metrics.

The financial data gives a clear picture of how Nucor’s cost structure has evolved. Net Sales and COGS move almost in parallel, which is typical for steel manufacturing where production costs represent the vast majority of total sales. Across the sample, the **COGS Ratio** remains very high (generally between 87% and 96%), confirming that even small changes in input prices have immediate effects on profitability.

Gross Margin performance improves during periods of strong demand or lower material prices—most clearly in 2021 (12.68%)—but tightens again when input costs rise, as seen during the inflationary pressures of 2022–2024. The data reinforces the idea that margin performance at Nucor is largely driven by cost behavior rather than revenue swings alone, which makes anticipating future cost movements particularly important for planning.

Inventory Turnover adds another useful perspective, connecting production cost forecasts with **asset use**. Years with higher turnover (e.g., 2016 or 2025 YTD) often correspond to stronger market conditions and more efficient use of working capital. Lower turnover values usually reflect slower demand or a deliberate increase in inventory levels to hedge against supply chain risks. Because steel production requires substantial investment in raw materials, turnover helps connect cost forecasting with balance-sheet efficiency and short-term liquidity management.

Table 3: Nucor Annual Financial Summary (2015–2025). 2025 values represent year-to-date totals (Q1-Q3), while ratios are averages for the period.

Year	Net Sales (\$M)	COGS (\$M)	COGS Ratio (%)	Gross Margin (%)	Inv. Turnover Ratio (x)
2015	12,190	11,680	95.82	4.18	1.60
2016	12,510	11,880	94.96	5.04	1.67
2017	15,470	14,290	92.37	7.63	1.44
2018	20,180	18,330	90.83	9.17	1.37
2019	18,540	17,070	92.07	7.93	1.17
2020	14,840	14,040	94.61	5.39	1.36
2021	27,050	23,620	87.32	12.68	1.36
2022	32,950	29,360	89.10	10.90	1.28
2023	27,000	24,290	89.96	10.04	1.29
2024	26,390	24,050	91.13	8.87	1.21
2025	24,851	22,590	90.90	9.10	1.37

5.5 Standardized Deviations and Outlier Detection

A Z-score-based analysis enables cross-variable comparison independent of scale, transforming each series to mean = 0 and standard deviation = 1:

$$Z_i = \frac{X_i - \mu}{\sigma} \quad (1)$$

The standardized view in Figure 6 reveals the statistical significance of observed deviations.

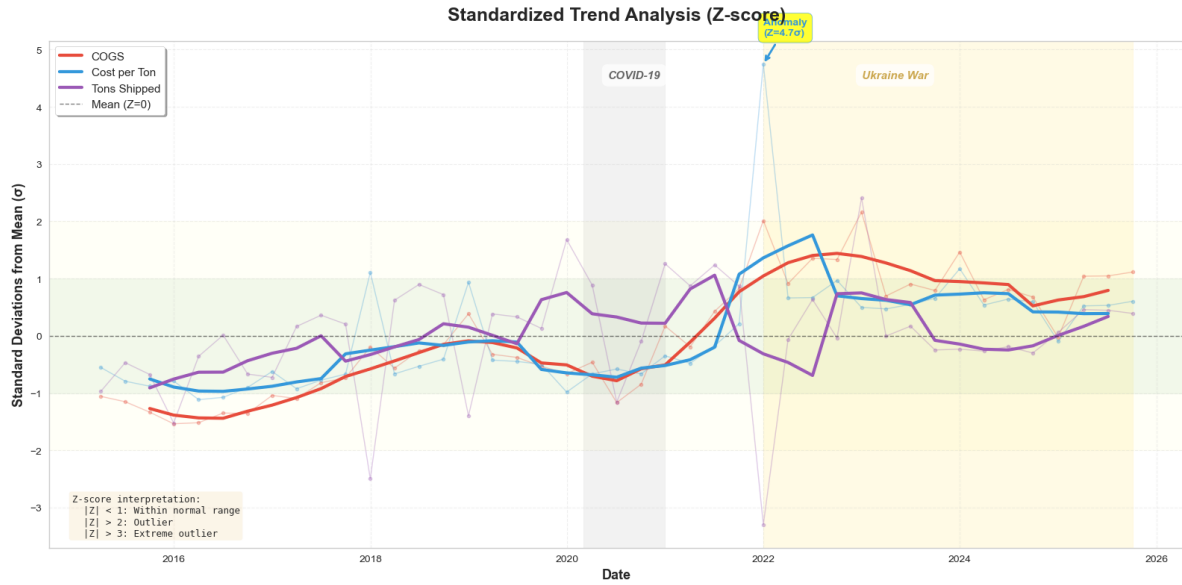


Figure 6: Standardized Trend Analysis (Z-scores). All variables transformed to standard normal scale.

The 2022 surge in production costs appears as an **extreme outlier** ($Z \approx 4.8$), representing a deviation exceeding 4.8 standard deviations from the historical mean. Under a normal distribution assumption, such an event would occur with probability less than 0.0001% (once every 250,000 quarters), clearly indicating a structural break rather than random variation.

Notable deviations are:

- **COVID-19 volume shock (2020 Q2):** Tons Shipped at $Z = -1.8$, representing a 1.8 standard deviation decline

- **Post-pandemic cost surge (2021 Q4):** Cost per Ton at $Z = +1.9$, reflecting supply chain bottlenecks
- **Ukraine War volume collapse (2022 Q1):** Tons Shipped at $Z = -3.2$, indicating severe demand contraction

COGS displays milder but still noticeable deviations in early 2023, signalling short-term operational distortions that gradually normalize.

5.6 Cost–Volume Relationship Analysis

Scatterplot analysis (Figure 7) confirms the negative association between unit cost and output volume, with clear visual clustering by time period.

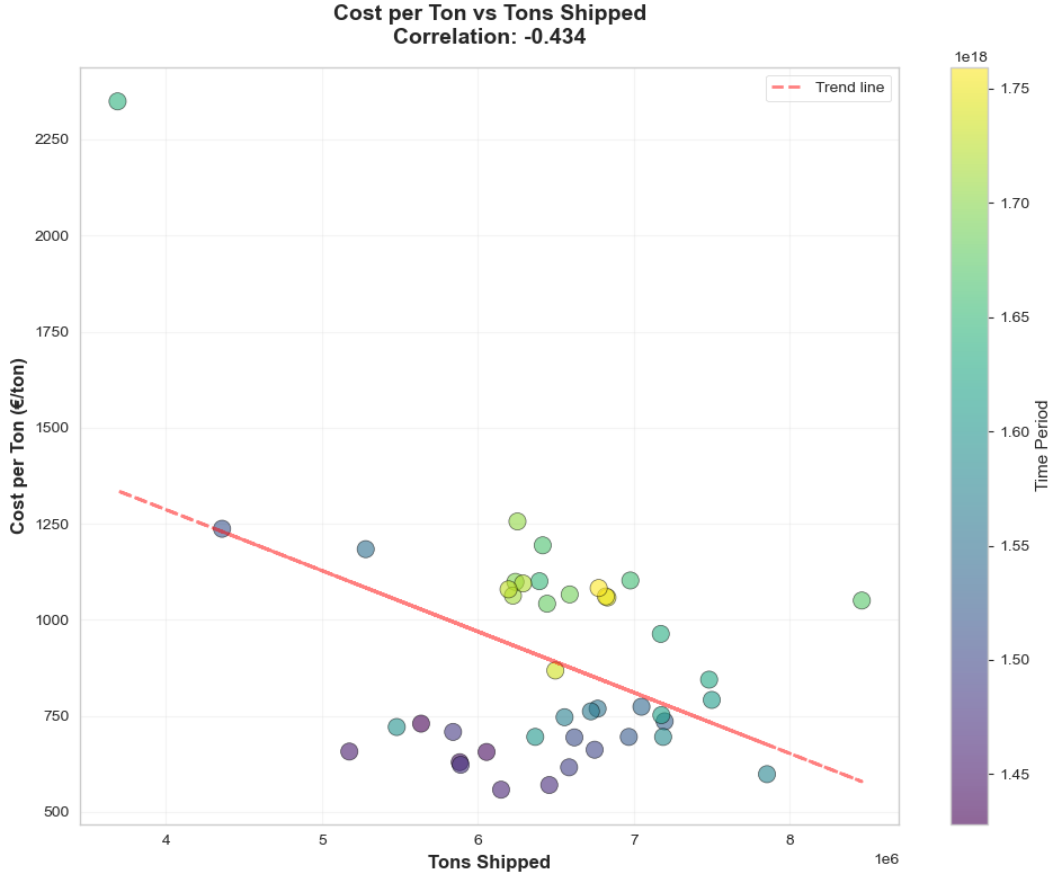


Figure 7: Cost per Ton vs. Tons Shipped Scatterplot. Each point represents one quarter, colored by time period (darker colors indicate more recent observations).

The downward trend line (slope = -89.3 \$/ton per million tons) indicates that each additional million tons of output associates with approximately \$89 reduction in cost per ton. However, the moderate $R^2 = 0.19$ suggests this relationship explains only 19% of cost variance, with the remainder attributable to external factors. This low explanatory power of output alone is precisely why our forecasting models incorporate macroeconomic features independent of the company, such as electricity prices, scrap prices, and commodity indices.

This pattern is consistent with three economic mechanisms:

1. **Fixed cost spreading:** Higher output distributes fixed costs (overhead, administrative expenses) over more units
2. **Operational efficiency:** Larger production runs reduce per-unit setup costs and improve equipment utilization
3. **Purchasing power:** Higher volumes may enable bulk purchasing discounts for raw materials

5.7 Rolling Correlation Dynamics

A 6-quarter rolling-window correlation (Figure 8) reveals that the link between shipments and unit costs is **not temporally stable**, violating the stationarity assumption required by many forecasting models.

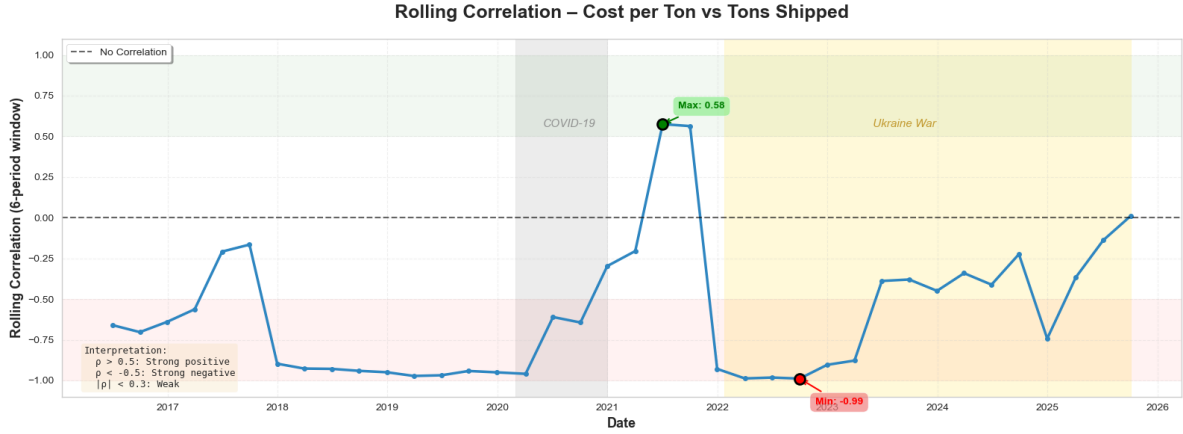


Figure 8: Rolling Correlation (6-quarter window) between Cost per Ton and Tons Shipped. Correlation coefficient computed using Pearson method over sliding windows.

The correlation exhibits three distinct regimes:

Table 4: Rolling Correlation Regimes and Economic Interpretation

Period	Mean ρ	Interpretation
Pre-COVID (2017–2019)	-0.68	Strong negative: Normal economies of scale operate; higher volumes reduce unit costs
COVID-19 (2020)	-0.92	Extreme negative: Fixed costs dominate; production cuts amplify per-unit costs
Recovery (2021–early 2022)	+0.54	Positive anomaly: Supply constraints invert relationship; higher demand increases both volumes and costs
Ukraine War (mid-2022–2023)	-0.78	Negative restoration: Demand destruction; high costs reduce volumes
Normalization (2024–present)	-0.42	Moderate negative: Return to historical pattern as markets stabilize

These patterns motivate the feature engineering and modeling strategies described in subsequent sections, particularly the need for regularization techniques and outlier-robust estimation methods.

6 Feature Engineering

The feature engineering phase aimed to transform raw financial and macroeconomic indicators into predictive signals capable of capturing the momentum and volatility of Nucor’s cost structure. Given the strong temporal dependence inherent in commodity markets, standard quarterly values were insufficient; thus, we implemented a pipeline to generate lagged and rolling features.

6.1 Base Feature Transformation

To integrate heterogeneous external datasets into our quarterly framework, we applied the following aggregation logic:

- **Price and Index Series:** Daily and monthly variables (e.g., Scrap Price, Electricity, USD Index) were resampled using period-end averages to align with Nucor’s reporting cadence.
- **Event-Based Data:** Discrete occurrences, such as NOAA disaster counts, were aggregated via summation to quantify quarterly disruptive potential.
- **Inventory Turnover (X_{12}):** A key internal operational metric constructed from cost-of-goods-sold and inventory levels to proxy production efficiency.

6.2 Temporal Feature Construction

To expose market dynamics to our predictive models, we expanded the feature space with two types of derived variables:

- **Lagged Features (X_{t-k}):** We generated lags ($k = 1$) for key drivers to account for the delayed impact of market shocks. For instance, a spike in scrap export volumes often takes a quarter to affect domestic price availability.
- **Rolling Statistics (X_{roll}):** We computed rolling standard deviations (window=3 quarters) to capture market volatility regimes.

Prevention of Data Leakage: All rolling features were computed on shifted data to ensure that information from the target quarter t never leaks into the predictors.

6.3 Feature Selection Strategy

Due to the limited sample size of the master dataset ($N = 43$), using the full expanded feature set would lead to overfitting. We therefore tried to understand which features are more relevant in order to train our models. We then studied which are the most significant non lagged features, and the most significant ones among all (non lagged, lagged).

6.3.1 Expanded Feature: Correlation, Lagged Effects, and Rolling Dynamics

Correlation Heatmap (Baseline, Lagged, Rolling). We began with a full inspection of all available features $X_1, \dots, X_{13}$ by computing their Pearson correlation coefficients with the target variable $Y = \text{Cost_Per_Ton}$. To capture potential dynamic relationships, this analysis was extended to **lagged features** $\{X_{i,t-k}\}$ for lags $k = 0, 1, \dots, 12$, allowing us to assess delayed effects of each cost driver.

The result was a comprehensive heatmap in Figure 9 matrix displaying correlations among raw and lagged predictors versus the dependent variable. This enabled a direct visual comparison of different temporal transformations and helped highlight which features retained predictive strength.

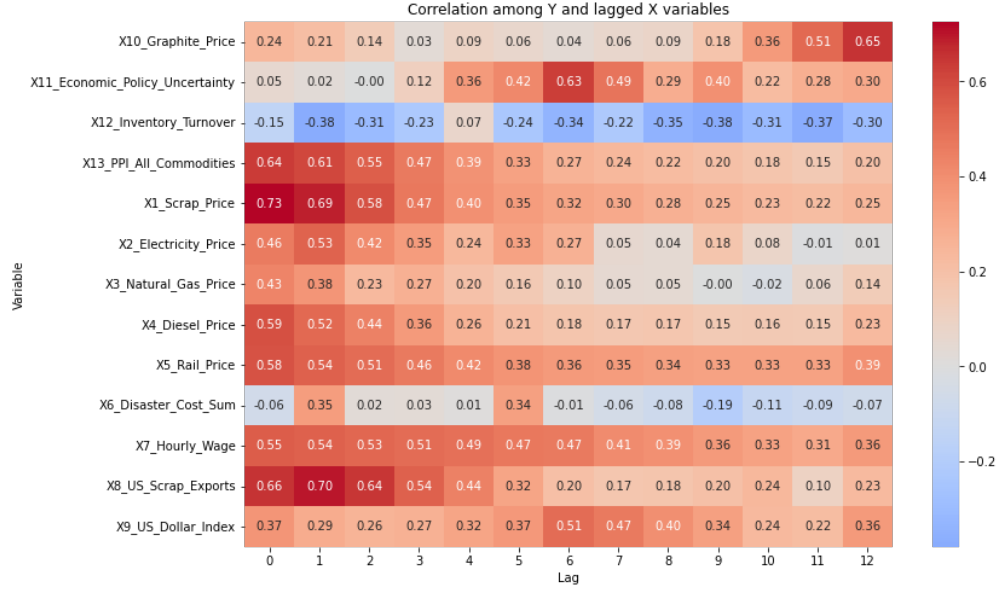


Figure 9: Correlation matrix between the target variable and lagged features.

Outcome of the Correlation Exploration.

The correlation structure revealed a clear hierarchy:

- **Scrap-driven and commodity-level variables** (notably X_1 , X_{8-lag1} , X_{13}) consistently displayed the strongest association with Y across raw and lagged forms.
- **Energy and logistics variables** (X_4 , X_5 , X_2 , X_3) showed moderate but persistent correlations, especially in lagged specifications.
- **Macro indicators** (X_9 , X_{10} , X_{11}) exhibited weak linear relationships, regardless of lag or smoothing.
- **Inventory and disaster-related variables** (X_{12} , X_6) showed weak or slightly negative correlations with Y .

Granger Causality

The Granger tests assess whether past values of each predictor improve the forecast of *Cost_Per_Ton* beyond the information already contained in its own history.

Strong evidence of predictive content. The following variables present extremely low p-values ($p < 0.05$), confirming they significantly help forecast cost:

- X_1 Scrap Price
- X_8 US Scrap Exports
- X_{13} PPI All Commodities

Second-layer operational drivers. Energy, logistics, labour, inventory efficiency, and disaster expenditures (X_2 , X_4 , X_5 , X_7 , X_{12} , X_6) show significant predictive content.

The remaining variables (X_3 , X_9 , X_{10} , X_{11}) do not have predictive power at lag 1.

Selected mixed set

To conclude, by taking into consideration both analysis, the mixed set of features that we identified was:

- **X_1 Scrap Price:** The baseline raw material cost.
- **X_8 US Scrap Exports (Lag 1):** Captures the delayed effect of international demand on domestic supply.
- **X_{13} PPI All Commodities:** A broad industrial inflation indicator.
- **Y Autoregressive Terms (Y_{t-1} , Y_{roll3}):** Capturing the inertial nature of production costs. These were added because past values of Y can provide useful information for predicting future values.

6.3.2 Statistical Feature Selection for the Non-Lagged Feature Set

To construct a second, fully non-lagged feature set, we complemented the correlation analysis with non-linear feature importance via Random Forests.

Random Forest Feature Importance

To capture potential non-linearities and interactions, a Random Forest model was trained on the non-lagged predictors. The ranking highlights:

- **X_4 Diesel Price — highest importance (≈ 0.28):** transportation and fuel shocks play a major role in explaining cost variability.
- **X_1 Scrap Price (≈ 0.18):** remains a primary cost driver, consistent with the correlation and Granger results.
- **X_8 US Scrap Exports (≈ 0.15):** confirms the global scrap tightness mechanism.
- **X_{12} Inventory Turnover (≈ 0.07):** operational efficiency matters in mitigating cost pressures.
- **X_3 Natural Gas Price (≈ 0.07):** gains relevance only when non-linear interactions are allowed.

Selected Non-Lagged Feature Set

Combining insights from correlations, Random Forest importance and validation through regression performance; our final non-lagged feature set is:

$$\left\{ \begin{array}{l} X1_Scrap_Price, X2_Electricity_Price, X4_Diesel_Price, \\ X5_Rail_Price, X7_Hourly_Wage, X8_Scrap_Exports, \\ X12_Inventory_Turnover, X13_PPI_All_Commodities \end{array} \right\}.$$

This set captures the core economic channels (scrap and global commodity pressure) together with energy and transportation costs, which play a significant role in the non-linear and operational dynamics of production costs.

6.4 Synthetic Data Augmentation

Given the limited number of historical observations, we generated an augmented dataset to assess model stability and convergence under a broader range of scenarios.

- **Methodology:** We employed a **Gaussian Copula Synthesizer** (from the `sdv` library). This approach learns the joint multivariate distribution of the original dataset, capturing the underlying correlation structure between the target variable (Y) and the feature set (X).
- **Output:** The synthesizer produced **200 artificial quarterly observations**, which were combined with the master dataset to create an expanded sample for model training.

- **Limitations:** The Gaussian Copula assumes that all observations are independent and identically distributed (i.i.d.). While it accurately preserves **static cross-sectional relationships** (e.g., the association between energy prices and Cost per Ton), it **does not retain temporal dependencies**. As a result, autocorrelation patterns, trend components, and shock-driven dynamics present in the historical time series are not reproduced in the synthetic data. This implies that models trained on the synthetic dataset are robust to structural value-based relationships but may underestimate temporal effects.

Because the synthetic dataset preserves cross-sectional correlations but **not time-series structure**, lagged and rolling features lose their economic meaning. For this reason, models trained on synthetic observations rely exclusively on the **non-lagged feature set** ($X_1, X_2, \dots, X_{13}$), focusing solely on structural relationships rather than dynamic behavior.

7 Modeling Approach

This section outlines the methodological framework adopted to predict Nucor’s quarterly **Cost per Ton** using both internal operational metrics and external macroeconomic drivers.

7.1 Feature Sets

As detailed in Section 6, the feature engineering stage produced two compact predictor configurations used throughout the modelling exercise. Here we briefly summarize their purpose.

1. **Non-lagged Feature Set (Baseline).** This set includes only contemporaneous macroeconomic and operational variables. It serves as a static benchmark capturing the immediate drivers of cost per ton.
2. **Mixed Lagged Feature Set (Engineered).** This configuration augments selected raw features with lagged terms and simple autoregressive components of the target. It is designed to incorporate the dynamic relationships identified in the correlation and Granger analyses.

7.2 Cross-Validation Strategy

Given the sequential nature of the dataset and the limited sample size, all regularized linear models (Ridge, Lasso, Elastic Net) and the Random Forest were evaluated using a **TimeSeriesSplit** cross-validation scheme rather than conventional shuffled folds. We adopted **5 splits**, which provides sufficiently large training windows and yields more stable estimates of the R^2 metric compared with higher-fold configurations.

This setup ensures that model assessment respects temporal ordering and avoids look-ahead bias while maintaining reliability despite the reduced dataset size.

7.3 Baseline Models

To rigorously evaluate the added value of machine learning, we established a non-parametric benchmark:

Naive Forecast (Persistence Model): This method assumes that the cost structure exhibits perfect inertia, predicting the next quarter’s cost simply as the current quarter’s value ($\hat{y}_{t+1} = y_t$). This serves as the “floor” for performance; any complex model failing to outperform the Naive baseline is deemed statistically insignificant.

7.4 Regularized Linear Models

Given the high multicollinearity among energy and commodity prices (e.g., correlations between oil, gas, and electricity), standard Ordinary Least Squares (OLS) estimates tend to be unstable with high variance. To mitigate this, we employed regularization techniques that constrain the magnitude of coefficient estimates.

7.4.1 Ridge Regression (L_2 Regularization)

Ridge regression adds a penalty term proportional to the square of the coefficients magnitude. This technique is particularly effective in our context because it shrinks correlated predictors (like various energy inputs) towards each other rather than discarding them, preserving the signal from the entire market basket. The objective function is minimized as:

$$\min_w ||y - Xw||_2^2 + \alpha ||w||_2^2$$

where α controls the regularization strength.

7.4.2 Lasso Regression (L_1 Regularization)

Lasso (Least Absolute Shrinkage and Selection Operator) penalizes the absolute value of coefficients. Unlike Ridge, Lasso can shrink coefficients exactly to zero, effectively performing automatic feature selection. This allows the model to isolate the few dominant cost drivers while zeroing out irrelevant noise.

7.5 Random Forest Regression

Random Forests are ensemble learning methods that construct a large collection of decision trees, each trained on a bootstrap sample of the data. Predictions are obtained by averaging across trees, which reduces variance and mitigates the instability typical of individual decision trees. For this project, Random Forests serve as a non-linear, interaction-aware benchmark: they can capture complex relationships among macroeconomic and operational variables without requiring explicit feature engineering or parametric assumptions.

7.6 Experimental: Deep Learning Feasibility

To explore the potential benefits of non-linear sequence modelling, we additionally implemented a set of deep learning architectures commonly used for time-series forecasting: Long Short-Term Memory (LSTM) networks, Gated Recurrent Units (GRU), and 1D Convolutional Neural Networks (CNNs). These models are designed to capture complex temporal dependencies through different mechanisms: recurrent gating (LSTM/GRU) and local pattern extraction (CNN-1D). Their inclusion serves as an upper-bound feasibility test, assessing whether richer temporal structure can be exploited beyond the linear and tree-based frameworks discussed earlier.

8 Results

This section presents the empirical performance of our forecasting models.

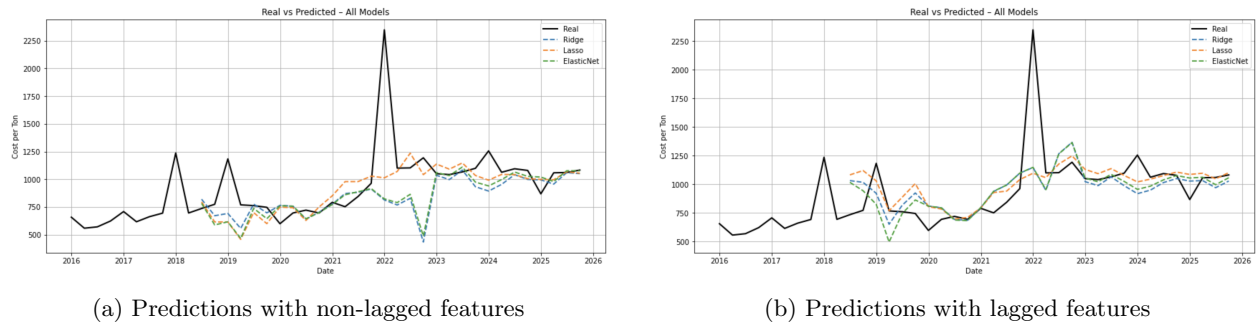


Figure 10: Comparison of Ridge, Lasso, and Elastic Net predictions using (a) non-lagged and (b) lagged feature sets.

8.1 Performance on the Original Dataset

We first evaluated Ridge, Lasso, and Elastic Net using a smaller subset of non-lagged features from the one identified in Section 6. The variables we picked (to avoid excessive overfitting) were:

$$\{X1_Scrap_Price, X8_Scrap_Exports, X4_Diesel_Price, X5_Rail_Price, X13_PPI_All_Commodities\}.$$

Cross-validation is performed with 5 folds under a TimeSeriesSplit scheme.

The average predictive performance across folds is:

$$\text{Mean } R^2_{\text{Ridge}} = -1.5757, \quad \text{Mean } R^2_{\text{Lasso}} = -1.4428, \quad \text{Mean } R^2_{\text{ElasticNet}} = -1.3853.$$

All models yield negative R^2 , indicating poor out-of-sample accuracy. Given the extremely small dataset ($N = 43$), this is expected: the models generalize poorly despite regularization.

8.2 Effect of Lagged Features

To assess the impact of temporal information, we replace the baseline raw predictors with the set of engineered lagged features:

$$\{X_1, X_{8_lag1}, X_{13}, Y_{t-1}\}.$$

The average cross-validated performance becomes:

$$\text{Mean } R^2_{\text{Ridge}} = -1.5155, \quad \text{Mean } R^2_{\text{Lasso}} = -1.1538, \quad \text{Mean } R^2_{\text{ElasticNet}} = -1.3496.$$

Two patterns emerge:

- Although the absolute performance remains poor, Lasso is the only model exhibiting some sensitivity to the inclusion of dynamic inputs.
- Ridge and Elastic Net remain comparatively stable, with performance similar to the non-lagged configuration.

Overall, lagged features do not drastically change the average performance, but they modify how models respond to periods of structural change.

8.3 Stress Test: Ukraine-War Volatility Window

To examine whether the lagged structure helps during turbulent periods, we inspected the most volatile segment of the series:

$$2021-06-30 \text{ to } 2022-09-30.$$

Ridge regression illustrates the contrast clearly:

$$R^2_{\text{Ridge, non-lagged}} = -1.0831 \quad \text{vs.} \quad R^2_{\text{Ridge, lagged}} = -0.0423.$$

The lagged specification achieves a near-zero R^2 , which is the best Ridge fold overall. This indicates that during the post-COVID and Ukraine-war regime—characterized by unusual volatility—lagged variables help capture short-run adjustment dynamics that the static feature set misses.

This can be intuitively seen by looking at the considered period in figures 10a and 10b.

Replacing Y_{t-1} with a smoothed autoregressive term Y_{roll3} slightly worsens the mean cross-validated performance (i.e., $\text{Mean } R^2_{\text{Ridge}} = -2.6695$), yet it produces a *positive* R^2 in the stress window:

$$R^2_{\text{Ridge, roll3}} = 0.2544.$$

Thus, lagged formulations do not consistently improve global performance, but they offer clear benefits during the analyzed structural break. However, this effect is context-specific, and, due to the small size of our dataset, cannot be generalized to other periods or datasets.

8.4 Synthetic Data Experiment (Gaussian Copula)

To investigate how model performance scales with larger datasets, we generated a synthetic dataset using a Gaussian copula fitted to the joint distribution of the predictors. The synthetic dataset was then combined with the available real observations to form a single, extended dataset. Cross-validation was performed on this combined dataset, allowing the models to learn from both real and synthetic data. Because the Gaussian copula does not explicitly preserve temporal dependencies, lagged features were not used in this experiment.

With an expanded synthetic dataset, model accuracy increases markedly (See Figure 11):

$$\text{Mean } R^2_{\text{Ridge}} = 0.4605, \quad \text{Mean } R^2_{\text{Lasso}} = 0.4617, \quad \text{Mean } R^2_{\text{ElasticNet}} = 0.4582.$$

This confirms that the poor performance on real data is largely driven by sample size limitations: with more observations, regularized linear models stabilize and achieve positive predictive power. However, these synthetic results must be interpreted with caution: Gaussian copula sampling disrupts the temporal structure of the series, reducing the experiment to a cross-sectional approximation rather than a true forecasting exercise.

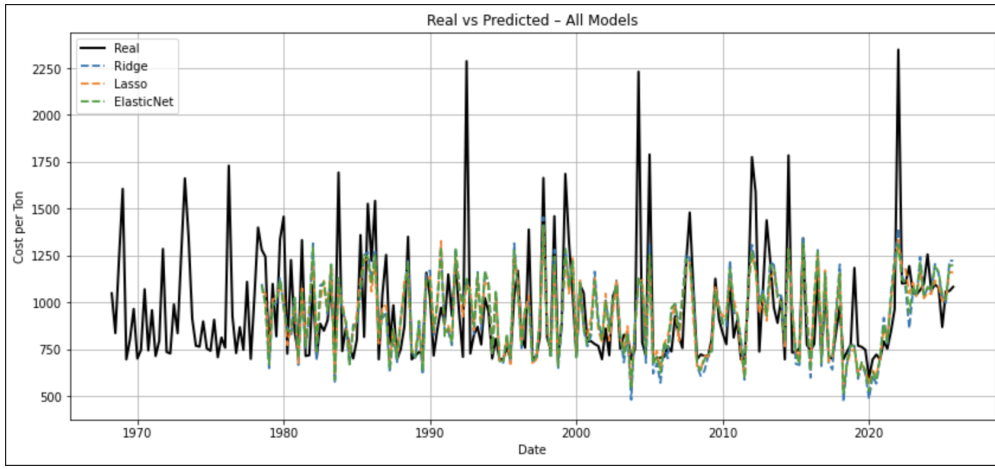


Figure 11: Predictions on synthetic dataset.

Summary of Findings

Across all experiments, three conclusions emerge:

1. With only 43 observations, all models struggle to generalize, yielding negative R^2 in true out-of-sample forecasting. In this scenario, Lasso seems the best model among the ones tested.
2. Lagged features do not uniformly improve performance, and the benefits found during high-volatility periods need to be further studied.
3. When supplied with larger datasets (with synthetic and real data), all models achieve positive and stable performance, confirming that data scarcity—not model choice—is the primary bottleneck.

8.5 Models Benchmarking with Synthetic Augmentation

Building on the results of the real dataset analysis, where **Lasso** was found to be the most stable regularized model under TimeSeriesSplit validation, we conducted a final benchmark comparing **Lasso**, **Ridge**, **Random Forest**, and a **naïve forecasting rule**.

In this experiment, cross-validation was performed exclusively on the real dataset using a walk-forward scheme. For each training fold, all synthetic data were added to augment the training set, increasing the effective sample size and providing additional variability for the models. Evaluation, however, was



Figure 12: Comparison of out-of-sample MAE across models under walk-forward forecasting. All models were trained on real data augmented with Gaussian Copula synthetic observations; evaluation is performed only on real future data.

always carried out on the held-out real observations, ensuring that performance metrics reflect the ability to generalize to actual data.

8.5.1 Naive Benchmark

As a baseline, we considered a naive forecast in which next-quarter cost is predicted to be equal to the previous quarter. This model establishes a minimal standard that any learning algorithm must surpass to be practically useful.

$$\text{MAE}_{\text{naive}} = \$165.78 \text{ per ton.}$$

8.5.2 Model Performance

Figure 12 summarizes the out-of-sample MAE obtained in the final experiment:

- **Random Forest:** MAE = \$185.86
- **Ridge:** MAE = \$134.70
- **Lasso:** MAE = \$134.28
- **Naive Benchmark:** MAE = \$165.78

8.5.3 Interpretation

The bar chart highlights three central findings:

1. **Random Forest underperforms the naive rule.** Its MAE (\$185.86/ton) exceeds the naive benchmark, indicating that a flexible non-linear model tends to overfit when the available real data are limited and structurally volatile. This is consistent with the instability already observed in earlier cross-validation experiments.
2. **Ridge and Lasso clearly outperform both the naive benchmark and the Random Forest.** With MAE values around \$134–\$135 per ton, both regularized linear models improve upon the naive baseline by approximately \$30–\$32 per ton. Lasso maintains a marginal advantage, in line with the previous results.

3. **Synthetic data help, but do not fully resolve the forecasting difficulty.** The introduction of Gaussian Copula synthetic observations stabilizes the regression estimates and reduces systematic error, but the remaining MAE indicates that the predictability of cost-per-ton remains limited. This is expected, as the copula generation breaks temporal dependencies and cannot replicate structural shocks (e.g., the 2021–2022 volatility regime).

Overall, these results suggest that in a **small-sample, high-volatility environment, regularized linear models (especially Lasso)** represent the most reliable forecasting approach. However, given the residual error magnitude, the models should be interpreted as *decision-support tools* for scenario exploration rather than as precise quarter-ahead predictors.

8.6 Results of Neural Network Models

To explore whether more expressive models could extract temporal patterns beyond the reach of linear methods, we conducted an additional experiment using three neural architectures commonly employed in time-series forecasting: **Long Short-Term Memory (LSTM) networks**, **Gated Recurrent Units (GRU)**, and **1D Convolutional Neural Networks (CNN-1D)**. All models were trained on a sliding-window dataset of length four quarters, using the full set of scaled engineered features. The data were split into training (70%), validation (15%), and testing (15%) following a chronological partitioning to preserve temporal ordering.

Table 5 reports the performance achieved by the three deep learning models across training, validation, and held-out test sets. For clarity, we summarize the key findings for each architecture below.

Table 5: Performance of Deep Learning Models (RMSE, MAE, R^2)

Model	Split	RMSE	MAE	R^2
LSTM	Train	0.577	0.349	0.742
	Validation	0.208	0.158	0.312
	Test	0.235	0.177	0.130
GRU	Train	0.661	0.392	0.661
	Validation	0.257	0.180	-0.056
	Test	0.252	0.195	-0.002
CNN-1D	Train	0.938	0.445	0.317
	Validation	0.443	0.359	-2.134
	Test	0.240	0.176	0.096

LSTM

The LSTM achieves the most stable results among the three architectures, with positive R^2 on both the validation and test sets. However, the gap between train and validation metrics indicates moderate overfitting, and absolute performance remains inferior to the Ridge and Lasso solutions.

GRU

The GRU exhibits clearer overfitting: good training performance but negative R^2 on both validation and test sets. The model fails to generalize, suggesting that the available data do not provide sufficient temporal information for recurrent architectures.

CNN-1D

The CNN-1D model shows unstable behavior, with a validation R^2 of -2.134 , one of the worst results in the entire study. However the test MAE is comparable to the LSTM.

Overall Assessment

Overall, the deep learning architectures exhibit clear signs of overfitting and instability. With only 43 quarterly observations and a temporal window of four periods, the models lack sufficient sequential depth to reliably learn generalizable temporal patterns. All three architectures, LSTM, GRU, and CNN-1D; fit the training data reasonably well, but validation performance falls quickly, reflecting the combined effects of small sample size and the high volatility of the series. Periods characterized by structural shocks are particularly challenging, and models struggle in capturing extreme fluctuations and non-stationary dynamics with limited data and complex architectures.

These findings confirm that in *small-sample, high-volatility environments*, complex neural architectures offer no systematic advantage over simpler, more robust methods (Lasso and Ridge).

9 Conclusions

This study aimed to develop a forecasting framework for steel production costs in an environment defined by extreme volatility, limited data availability, and abrupt structural breaks. By comparing baseline methods, regularized linear models, and deep learning architectures, several clear conclusions emerge.

9.1 Key Findings

1. Simplicity Outperforms Complexity

Complex non-linear models did not generalize effectively in this small-sample, high-volatility setting. Random Forest underperformed the naive benchmark, while deep learning architectures (LSTM, GRU, CNN-1D) exhibited severe overfitting, yielding negative or near-zero R^2 on validation and test sets despite fitting the training data.

Regularized linear models (Lasso, Ridge) proved far more stable and achieved the strongest out-of-sample performance, improving upon the naive baseline by approximately 19%.

2. The Role of Synthetic Data Augmentation

Models trained solely on the real dataset initially yielded negative R^2 , reflecting insufficient data for reliable learning. Synthetic augmentation via Gaussian Copula significantly stabilized model behavior, increasing cross-validated R^2 from approximately -1.4 to $+0.46$.

However, copula-based generation breaks the temporal structure of the series, effectively reducing the experiment to a cross-sectional approximation. Consequently, models trained on synthetic data cannot capture high-frequency shocks with precision. Even in the final benchmark, the residual MAE ($\sim \$134$) highlights the inherent difficulty of quarter-ahead forecasting in turbulent markets. Synthetic data therefore serves primarily as a tool for scenario exploration rather than precise forecasting, yet remains a necessary precondition for machine learning in low-frequency industrial datasets.

3. Regime-Dependent Role of Lagged Features

Lagged features showed some improvement in validation performance during the 2021–2022 Ukraine-war window, suggesting that short-term momentum may carry information under extreme volatility. However, given the very limited size of our 43-quarter dataset, this effect is context-specific and cannot be generalized to other periods or crises. While lagged features appear to contribute meaningfully in certain scenarios, their predictive value should be interpreted with caution and treated as non-negligible complementary to raw macroeconomic drivers.

9.2 Strategic Summary

The study demonstrates that the steel market underwent a structural break after 2021, making traditional historical averaging ineffective. Empirical results show:

- Deep learning models are unsuitable under quarterly data scarcity.

- Regularized regression provides the best balance between accuracy, robustness, and interpretability.
- Forecasting abrupt market shocks remains the principal limitation.

Despite these constraints, the forecasting framework offers strategic value for Nucor:

- **Hedge Raw Materials:** Anticipate cost uptrends and secure scrap contracts accordingly.
- **Dynamic Pricing:** Adjust steel prices in advance of predicted cost increases.
- **Operational Agility:** Manage inventory turnover and production scheduling to mitigate volatility exposure.

These insights position *Cost Foresight* as a valuable decision-support system for steel cost management, capable of guiding medium-term planning even if not designed for precise quarter-ahead shock prediction.

9.3 Benefits Analysis

The project yields several practical benefits for steel cost management:

1. Understanding of Cost Drivers

Integrating internal financial metrics with external macroeconomic indicators clarifies the variables that most strongly influence cost per ton, particularly scrap prices, energy costs, logistics, and inventory dynamics.

2. Insights into Temporal Effects

Although their benefits require further investigation, lagged features may provide modest improvements, highlighting the importance of capturing short-term momentum and inventory-driven dynamics.

3. Foundation for Future Modelling Improvements

Although the forecasting performance remains limited, the project establishes a rigorous methodological foundation on which more advanced models can be developed. This includes a validated feature framework and a clear understanding of which modelling strategies are viable. These elements provide a solid starting point for future research.

9.4 Future Research Directions

Future work may address limitations identified in the current study:

- **Time-Series Generative Models**

Since Gaussian Copulas break temporal dependencies, future research may explore **Time-Series GANs** [10] or **time-aware bootstrapping** [11] to generate synthetic data that preserves autocorrelation structures.

- **Regime-Switching and Adaptive Models**

Given the regime-dependent value of lagged predictors, exploring models such as regime-switching architectures, adaptive windowing, or Markov-switching dynamics may improve responsiveness during structural breaks.

- **Higher-Frequency External Data**

Integrating weekly or daily scrap prices, energy spot prices, and freight indices could bridge the gap between real-time market conditions and quarterly financial reporting.

- **Industrial Integration of High-Frequency Internal Data**

Since quarterly aggregates limit model capability, accessing Nucor’s internal ERP or SAP systems would enable daily-level modeling. This would alleviate the small-sample constraint and make advanced architectures (LSTM/GRU) viable and potentially superior.

- **Ensemble and Hybrid Approaches**

Combining the stability of regularized linear models with the interpretability of domain-driven feature engineering and the flexibility of machine learning may yield more robust forecasting tools.

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